

CARRY-CONVEXITY: Pricing the Optionality the BuckBasket Sells

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1 Purpose and lineage

This document joins the excursion-catalogue program (REBALANCING-EQ.org) with an external critique: Panoptic's "Yin (Carry) & Yang (Convexity) Require Each Other to Survive & Thrive" (X article, 2026-08-27; saved copies `~/Downloads/Panoptic-carry.{pdf,html}`), itself the onchain transplant of the Artemis/Cole carry-vs-convexity framing. The critique, stated in our terms:

A passive AMM LP is a carry trader – a seller of perpetual optionality who collects a fixed fee (the implied volatility) while eating adverse selection (the realized volatility). Onchain, the structural oversupply of LP positions suppresses IV below RV, and crypto is the most momentum-prone, fat-tailed asset class – so the profitable side is BUYING convexity from LPs, not being one. Carry books show negative skew ("sawtooth" P&L: steady fees, rare wipeouts); convexity books show positive skew ("stairstep"). A market is complete and stable only when both sides exist and both are PRICED.

The critique matters here because the BuckBasket is not merely **an** LP – it is the **SOLE** full-range LP in every TOKEN/BUCK constituent pool (`alberta_buck/sim/deploy.py` `pool_meta`; `BuckBasketUniswapV3.sol`), its depositor-facing balance sheet IS that LP book, and `basketValueInBuck` (the controller's process variable) is read from those same pools. Every arb, whale slice and excursion trade in a constituent pool has the basket (depositor L + treasury L) as counterparty. There is no third party absorbing adverse selection.

Three goals organize the program (decisions of intent, not yet of record):

1. **Harvest the benign workforce.** The BUCK-unaware arbs (AnonymousArb, TokenAccumulator lineage) repin the TOKEN/BUCK pools to the world for free and pay the basket fees to do it. Their flow must STAY cheap – the fee is "the gate that decides how small a mispricing an external arb will bother to close" (`deploy.py` fee note) – while the basket captures it as carry.
2. **Detect adverse events and price them.** Toxic flow (informed, manipulative, raid-speed) must pay a higher price for the convexity it

consumes, detected from observables (markout, tick RV, |bvib-1|, size vs impact norms) and priced via state-dependent fees and spreads.

3. **Position ahead of the system's own enforcement.** K's trajectory is committed, slow and public (integral-dominant PID, $\tau_I \sim 90$ d). The basket can legitimately place inventory, ranges and fees aligned with what the controller will enforce – the common-mode analogue of the fs30 foresight result, available to the system for free.

2 The audit: where the critique lands (2026-09-01)

2.1 The exposure

- The basket owns one full-range V3 position per constituent, in that constituent's TOKEN/BUCK pool; economically a CPMM wearing a V3 interface ($NAV = 2 \times \text{buckReserve identity}$, `BASKET-REDESIGN.md`). Full-range = the widest possible short straddle, sold continuously at the pool fee, at every price.
- Fee tiers are static: TOKEN/BUCK 3000 (0.30%), BUCK/USDC 500. One IV for all regimes and all counterparties.
- The depositor is the residual claimant on BOTH commodity moves and BUCK's par deviation (`BuckBasketProRata.sol _redeem` phase 2: BUCK profit $\rightarrow$ treasury; shortfall $\rightarrow$ covered from depositor TOKEN). The treasury holds a one-sided call; the depositor holds the negative-skew residual, down to the **Underwater** revert as the left tail.

2.2 The measured evidence (already in this repo, not previously named)

- `alberta_buck/sim/plot_basket_split.py` (docstring analysis): one agent pushing BUCK $\sim 19\%$ off parity made EVERY redemption take the deflation branch; $\sim \$1.54\text{M}$ over ~ 300 redemptions landed on depositors "rather than on the agent causing it." That is the Panoptic asymmetry, measured.
- Finding 6 (`REBALANCING-EQ.org`): the MA-reversion differential rebalancer loses $\sim 6\%$ of inventory per year trading trending macro drift as if it were reversion – $RV > IV$ on the differential axis, measured.

- Finding 11: foresight_days=30 flips the same book to +\$4-5M – the informed convexity buyer’s profit is real, and the passive quoting side pays it.
- alberta-buck-operations.org ops-desk scoreboard: the basket desk loses on drift (2/9 seeds profitable) – exactly where the move is real – and "a lagged operator is a profitable destabilizer."
- test/basket/BuckBasketFence.t.sol two-leg test: "the basket takes genuine impermanent loss that swamps the fee income being measured."
- LVR / markout accounting is absent. snapshot.py LP APR is gross fee yield with no adverse-selection offset. The repo currently cannot answer: did the 30 bp fee on NRG/BUCK cover realized vol last quarter?

2.3 Where the critique does NOT land: manufactured reversion

Panoptic’s $RV > IV$ inversion is an empirical claim about UNANCHORED, momentum-prone assets. The bvib axis is engineered to mean-revert: K plus the poised cast manufacture the reversion that a TradFi option seller merely bets on. The catalogue proved it: attacker P&L is superlinearly negative (-\$0.87M / -\$3.06M / -\$9.13M at x1/x2/x4 squeezes), defenders are paid countercyclically (finding 8), and the excursion agents’ min_edge gate means they refuse to sell convexity below a 1% floor premium – priced optionality, not passive LPing.

So the two axes split cleanly:

axis	vol regime	correct posture
common mode (BUCK vs basket)	manufactured reversion	SELL convexity – but priced, never
differential (TOKEN i vs j)	genuinely trending (macro)	patience (director), paid foresight, l

The prescription is not "stop being the carry side." It is: price the convexity currently given away (goals 1-2), route carry income to the standing convexity that defends the system, and buy the convexity the system cannot manufacture (the trending satellites). The insurance company that also owns the fire department should charge premiums, pay its firefighters a retainer, and reinsure the one block it cannot protect.

3 Wave 3 instrument: the markout / LVR ledger

The catalogue’s rule applies: no new machinery before the falsifiable instrument. Wave 3’s instrument is a per-pool, per-counterparty-class markout ledger.

3.1 Definitions

For each swap s in constituent pool i at time t , with the basket-LP side’s signed base-token flow q_{lp} and effective execution price p_{exec} :

$$\text{markout_D}(s) = q_{lp} \times (p_{ref}(t + D) - p_{exec}(t))$$

Negative markout: the basket sold what was about to appreciate (or bought what was about to fall) – adverse selection, realized. Three reference prices, in order of increasing externality:

- (a) own-pool TWAP at $t + D$ – cheap, self-referential;
- (b) truth-implied $\text{TOKEN/BUCK} = (\text{TOKEN/USDC}) / (\text{BUCK/USDC})$ from the deep truth pools – the world’s opinion, the one the whale cannot cheaply move; the default reference;
- (c) the exogenous reference series itself – the sim’s oracle truth, the fs30 analogue; sim-only, bounds what any detector could achieve.

Horizons D in {next tick, 1 d, 5 d}, matching the dual-timescale excursion signal. Aggregates per pool per window:

fees_i = fee income accrued to the basket position adverse_i = sum of markout_D over swaps (signed) $\text{netlp_i} = \text{fees_i} + \text{adverse_i}$ (the LP’s true P&L vs hold) $\text{lvr_bench_i} = \text{integral of } (\sigma_i^2 / 8) \times \text{poolValue_i} \text{ dt}$ (analytic LVR from realized tick variance) $\text{carry_ratio_i} = \text{fees_i} / \text{lvr_bench_i}$ (the IV-vs-RV verdict)

3.2 Attribution: the sim’s unique advantage

Every sim swap has a known sender class (keyed RNG, fixed roster), so the ledger splits per class x pool x window: whale (accumulate / dump / re-buy phases separately), BUCK-unaware arbs, director keeper, excursion agents by base, DM depositors, debtors. Toxicity of a class = mean markout per dollar of its flow. Expected shape, which the instrument must confirm or refute:

- whale dump-day flow: deeply negative for the basket (the attack IS concentrated adverse selection);
- benign arbs: mildly negative gross, $\sim$ zero net of fee (they sell the basket exactly the rebalancing it wants – Shannon’s demon’s other half);
- director keeper: the basket trading against its own book (internal transfer; must net to $\sim$ fees);
- excursion agents: negative for the pool ledger BY DESIGN (they trade only at $\geq 1\%$ edge) – the defense is partly funded by the basket’s own LP book, a transfer the system-level P&L hides and this ledger exposes.

3.3 Where it lives

- `snapshot.py` already reads fee growth (`feeGrowthGlobal` deltas + `tokensOwed`); add swap-event capture with class tags and the deferred markout resolution at frame `t + D`.
- `eqmetrics.py`: a markout panel alongside `excursion_windows` – per-pool fees / adverse / netlp / `carry_ratio`, per-class toxicity, for the whole run and inside each injection window.
- `catalogue.py` grid columns: basket fees, basket adverse, netlp, `carry_ratio` (worst pool), whale toxicity extracted.

3.4 New gates (additive to the 2-year tail gate)

1. **Carry coverage:** `tail carry_ratio_i` ≥ 1 for every constituent pool – the fee actually covers the realized-vol cost of the depth sold.
2. **Benign-flow health:** quiet-regime (outside injection windows) tracking error of each pool vs the truth-implied price stays within the current baseline $\times 1.25$ – whatever fee policy is adopted must not starve the workforce that pins the sensor.
3. **Expropriation bound:** depositor deviation-branch cost per dollar of injected manipulation flow (the `plot_basket_split` decomposition) must FALL under any adopted option, never rise.

4 Implementation options

Ranked lightest to heaviest. A/B/C are pool-structural; D is an overlay; E/F recur from the quadrant program and are listed for completeness. Everything is sim-testable before contract work: model the fee/structure in the pyrevm pool wrapper first (exactly as `fundingFactor` was simulated before on-chain enforcement), port the winner later.

4.1 A. Static fee retune (V3, no new machinery)

Let wave 3a's carry_ratio pick per-pool static tiers: plausibly cbBTC/BUCK and PAXG/BUCK up (trending, fat-tailed – the underpriced straddles), LABR/CNST down (slow anchors, benign flow dominates, cheap tracking matters more than premium). Cost: none. Limit: one IV per pool for all regimes; the one-day squeeze stays unpriced; known to be a compromise (`deploy.py` three-way fee tension note).

4.2 B. Uniswap v4 hooked pools: dynamic fees

V4 pools may declare a dynamic-fee flag; a hook contract sets the fee per-swap (beforeSwap), with afterSwap available for accounting. The basket already owns, bootstraps and registers its pools, so migrating a constituent to a hooked pool is an internal registry + LP move (the adoptable-receipt / constituent-migration scaffolding is the path; note current deps are v3-core/periphery – v4-core would be a new vendored dep, EVM target cancun already satisfied).

Candidate fee policies (the wave-3b axis):

policy	fee	prices
vol-scaled	$\text{base} + kv \times \text{EWMA}(\text{tick moves}^2)$	raid-speed realized vol
dev-scaled	$\text{base} + kd \times \text{abs}(\text{bvib} - 1)$	excursion persistence
directional	high on deviation-worsening side, base on the correcting side	the attack itself
size-tiered	convex in size / impact-cap norm	block toxicity (uncapped)

Properties worth having regardless of policy choice:

- Surge revenue accrues to the LP position = depositor L + treasury L: the manipulated are paid BY the manipulation, countercyclically – the fee-level analogue of finding 8, and the direct repair of the `plot_basket_split` expropriation.

- The directors already maintain tick-EMA ladders on these pools (`PairsRebalanceDirector`); the vol estimator is largely built.
- The one-day squeeze trough that no daily-gated defender touched (wave-2 conclusion) is exactly what a per-swap fee CAN touch: it is the only tick-speed responder that needs no agent, no capital and no signal faster than the trade it prices.
- am-AMM variant (auction-managed AMM): auction the right to lead-trade the pool each block/epoch; the rent converts LVR into income mechanically. Heavier, changes the pool's trust surface; catalogue it as a study behind the simpler fee policies.

Sim plan: implement the fee schedule as a state function in the sim's pool adapter (no v4 port needed to LEARN); gate any Solidity hook work on a wave-3b win.

4.3 C. Priced-tail liquidity: the fence generalized

Full-range custody sells optionality at every price for free. The fence already reframes depth as "a continuum of resting limit orders" in a +/-15% band. Generalize into three structural zones per pool:

1. a **band** around TWAP holding most depositor L (dense where benign flow lives – multiplies fee capture per dollar exactly where volume is);
2. **priced tails**: single-sided range orders at entry-grade offsets (3%+), which fill only at a charged discount/premium -- the `~min_edge` discipline expressed as position structure instead of an agent. A raid that dumps through the band fills the discount tail: the basket buys the dump at a price it set in advance, which is inventory acquisition at a premium, not silent IL;
3. an **idle reserve** (the fence's (1-K) buffer lineage) sized by K.

Wave-2 already showed the basket-rotation defender was the best scaler but capacity-limited by thin pools; the basket's own book restruct as band+tails is that same strategy at two orders of magnitude the size. Note what this changes for valuation: the $NAV = 2 \times \text{buckReserve}$ full-range identity dies with full-range; `poolBuckValues` must price the position properly (the fence already crossed this bridge).

4.4 D. Derivatives overlay: perpetuals and options

Three distinct legs; keep them separate in design and in accounting.

4.4.1 D1. Hedging the trending satellites (buy convexity)

The differential axis is genuinely trending and the basket is a structurally underpaid vol seller there (finding 6; the "deflation forever" trap, alberta-buck-ethereum-basket.org). The 600 bp cbBTC cap is the first defense; the second is to hold protective tail exposure on the satellites, bought externally and funded from the rebalancing premium (+322-496 bp/yr measured, `BASKET-REBALANCE.md`):

- venue options: perp futures on cbBTC/PAXG-proxies at external venues – Hyperliquid (CLOB perps; HIP-3 permits builder-deployed markets as of this writing), dYdX, GMX – or perpetual OPTIONS (Panoptic-style, on truth-side pairs) where available; options are the honest instrument (the exposure being hedged is gamma, not delta), perps the liquid one (a delta-hedge program on perps replicates convexity at rebalance cost – acceptable if sized small);
- sizing from the wave-3a ledger: hedge the measured adverse_i of the satellite pools, not a model;
- success metric: depositor IL on spike/step arms with hedge on/off, net of hedge cost (funding + fees). If the hedge does not beat its carry cost across x1.3 AND x1.6, the correct hedge is a smaller satellite weight, not a derivative.

Caveats: venue/counterparty risk (an external CLOB's margin engine and custody are outside our solvency argument), basis risk (cbBTC vs venue BTC-perp), and operational keys. The overlay must be a treasury program with its own budget and kill switch, never entangled with depositor L.

4.4.2 D2. Writing par-reversion insurance (sell convexity, explicitly)

The system's crown asset is manufactured reversion at par. Sell it explicitly instead of implicitly:

- form: a funding-rate perpetual on bvib (mark = medianized bvib TWAP). Deviation-longs (bets that $|bvib-1|$ grows) pay funding to the writer while inside the band; payout on excursions. The treasury is the natural writer because it enforces the reversion it is selling;

- the credit-base excursion agent already IS this trade implicitly (mint/sell above par, cover below); D2 formalizes it with explicit premium accounting – streamia for the backstop currently free;
- the solvency condition is the point: an attacker expressing the raid THROUGH the instrument (buy deviation-longs, then dump) must pay more in funding + premium + fee surge than the defended excursion can return. Wave 2 measured the defended excursion envelope (peak x duration at x1..x4); price the instrument off that envelope, cap open interest against the defense budget, and verify the superlinear-loss property survives listing. If it does not price safely, do not list: D2 is a study with an honest kill criterion, not a default.

4.4.3 D3. Perp telemetry as an early-warning sensor (detection)

Raids have a quiet phase (60-day accumulation before the dump). Off-pool venues see it first: funding rates, open interest and basis on BUCK and satellite pairs move while the accumulator is still impact-capped. Feed external funding/OI/basis into the adverse-event state that drives the fee policy (B) and arms the defenders. In-sim analogue now: the whale’s accumulation is visible in pool-imbalance and flow-toxicity terms – add a detector metric and measure LEAD TIME on the dump/squeeze arms (how many days before the dump does the detector fire, at what false- positive rate on the no-injection control?).

4.5 E. Auctioned rebalancing flow (RFQ) – recurring from the quadrant program

Findings 6/11: mechanistic differential rebalancing is negative-carry; foresight is a business. The basket cannot have commodity foresight but can auction its flow to those who do: director publishes intent (sell i, buy j, size, deadline); solvers bid; best execution wins. The basket stops paying LVR on its own rebalancing legs and captures part of the informed edge as price improvement. The stubbed `BasketRebalancer.pathFor` registry and the unmodelled keeper bounty both resolve into the auction spread. Sim: reuse the fs30 agent as a bidding solver vs the pool-walking baseline.

4.6 F. Readiness yield – carry funds convexity

The article’s deepest point applied to the quadrant program: convexity books die without carry. The poised cast bleeds demurrage while waiting (finding

5), Q3 overcrowds into losses at x4 (finding 10), and the x4 bound showed the cast's EXISTENCE is the difference between equilibrium and permanent failure. Route treasury income (BUCK-profit accrual, fee-surge revenue from B, funding from D2, funding-factor tolls) into a demurrage rebate for capital verifiably parked in defense posture (a vault whose funds can only trade toward par). With endog-style arrival hazards on defenders, the cast then SIZES ITSELF to the yield. Wave-3 question: the minimum readiness yield at which the x4 dump bound holds with an endogenous cast.

5 Policy foresight: positioning ahead of K

The unique, legitimate edge. Sign conventions first (decision of record): $bvib > 1$ = basket dear = BUCK cheap = DISCOUNT excursion; error = $1 - bvib < 0$; K falls; credit contracts; BUCK strengthens; $bvib$ returns to 1. Premium excursion mirror-image ($bvib < 1 \rightarrow K$ rises $\rightarrow$ supply grows $\rightarrow$ BUCK weakens back to par).

Because the controller is integral-dominant and slow ($\tau_I \sim 90$ d), its trajectory over the next weeks is a committed function of its readable state. During a sustained discount the system KNOWS contraction is coming and BUCK will be pushed up; during a sustained premium, the reverse. Three legitimate uses – this is not front-running external information, it is the central bank buying its own currency ahead of its own announced tightening:

1. **Inventory:** BuckBasketOps Q1 absorbs the discount (accumulate BUCK the policy is about to make scarce); Q3 supplies the premium. Both coincide directionally with the excursion agents' reversion trades – the gradient gate adds confirmation and timing, not direction.
2. **Ranges:** skew the priced tails (option C) toward the enforcement direction – deeper bid-side tails during a discount.
3. **Fees:** pre-emptively raise the anti-enforcement side's fee (option B directional policy keyed on the K gradient rather than raw $bvib$).

A reconciliation flag, raised rather than silently re-litigated: the roadmap ruling (REBALANCING-EQ.org quadrant-balance model) phrases the ops condition as "absorb . . . when BUCK is at a discount and K is RISING"; under the sign convention above a discount drives K DOWN (integral dominance makes dK/dt track $\text{sign}(\text{error})$ almost immediately), so the phrase cannot describe K's contemporaneous response to the same discount. A 2026-09-01

code audit found the tension lives ONLY in prose – every shipped signal is correctly signed (see the audit note below) – and that "K rising" has TWO coherent readings as a gate on the unimplemented BuckBasketOps Q1 condition:

- **Gate A – enforcement alignment (absorb while K is falling):** buy what policy is actively making scarce; the K forcing closes the position. Supported by `BuckPoolInvestorAgent` (which already places ranges on the side K is pushing toward, docstring stating the convention exactly) and by `BuckBasketOps.monetaryDeviationOffset`'s stranding analysis (the desk's position needs the K forcing behind it – "if K then stops tightening, the long-term forcing the desk's position is a bet on never arrives").
- **Gate B – deploy-only-when-healthy (absorb transient discounts against a rising TREND-K regime):** K read at regime scale ($\tau_I \sim 90$ d) classifies the discount – transient dip in an expansionary regime (safe to buy; the corner the slow lever cannot serve) vs structural oversupply in a K-falling regime (stand down; the controller is handling it, and the monetary book would be spent fighting a correction). Supported verbatim by the fence's established K-coupling (`BuckBasketFence.sol`: "K falling forces it to withdraw and burn; K rising lets it deploy") – the same design instinct for the same organ family.

The readings differ in timescale (instantaneous dK vs regime trend) and in objective (profit alignment vs book preservation), and both have design-text support. Wave 3d settles it empirically: gate variants {none, dK<0 fast, trend-K rising, composite} on the Q1/Q3 ops program, measuring both ops P&L and stability – per the ops-desk law ("profitability and stabilization ... come apart" only when the signal lags), and whichever gate wins IS the decision of record.

Audit note (2026-09-01): no implemented agent trades on an inverted signal. `ExcursionArb` and subclasses buy the discount / sell the premium / close on reversion (bvib EWMA only; K used solely as mint capacity); `SaverAgent` code is correct (`discount = max(0, bvib-1) -> buy`) though its inline comment at `alberta_buck/sim/equilibrium_agents.py:2099` states the convention INVERTED ("BUCK is discounted when ... `basketValueInBuck < 1`") – a stale prose slip contradicted by its own next line; the endog hazards recruit issuers when BUCK is rich (correct); and the shipped monetary path is correctly signed end-to-end (`PairsRebalanceDirector.monetaryEffort`: "`cm > 0 = BUCK CHEAP = retire (negative)`" -> `BuckBasketOps effortBp <`

0 -> `_absorb`), with NO K-gradient gate implemented anywhere yet. The catalogue results corroborate: a backwards cast would amplify excursions and bleed; the measured cast damps peaks and is paid more the worse the attack.

Two corollaries worth writing down:

- The controller’s state is public, so EVERYONE has policy foresight. That is a feature: third parties front-running committed policy do the enforcement for the system, so K itself has to move less (forward guidance). The basket’s edge is not informational – it is that it can position at zero adverse-selection cost against its own book and it earns the reversion it enforces.
- fs30 priced perfect 30-day COMMODITY knowledge at +\$4-5M on an \$8.6M book. Wave 3d prices perfect POLICY knowledge – which the system possesses for free – on the common mode. If the number is material, the ops program pays for the readiness yield (F) by itself.

6 Wave 3 cell plan

wave	axis	cells
3a	instrument only	existing arms x {none, usdc, all}; no new physics
3b	fee policy	{static-retune, vol, dev, directional, size} x {dump, squeeze, grind-down}
3c	tail structure	{full-range, band, band+tails} x raid arms (fence-in-the-loop, at last)
3d	K-foresight gates	Q1/Q3 ops x {none, dK<0, dK>0, predicted-K}
3e	derivatives overlay	D1 hedge on/off x {spike, step} x {x1.3, x1.6}; D2 writer solvency; D3 le

Gate set: the 2-year tail gate, plus carry coverage, benign-flow health, and the expropriation bound (wave-3 instrument section). 3a reuses the existing grid so every historical arm gains the new columns on regeneration; 3b-3e are new physics and go behind it.

7 Recommendations (ranked by expected information per unit of compute)

1. **Build 3a.** Pure metric; no model changes; reinterprets all 86 existing catalogue cells. Everything else in this document is hypothesis until `carry_ratio` and the toxicity ledger exist.

2. **3b in the sim's pool adapter.** The dynamic fee is the only candidate answer to the untouched one-day squeeze trough, and the sim can test every policy without touching Solidity.
3. **3c.** Fence-in-the-loop was already the matrix's recommendation 5; the markout metric finally gives it a depositor-economics judgment, not just a stability one.
4. **3d.** Settles the roadmap's K-gradient sign with data and prices the system's free policy foresight.
5. **3e.** Overlay studies; D1 first (bounded, honest kill criterion), D3 second (pure telemetry), D2 last (only if the solvency pricing closes).
6. **Contract work LAST:** vendored v4-core + a hook implementing the winning 3b policy, the band+tails custody rework if 3c wins, registry migration via the receipt-adoption scaffold. Nothing ships on-chain that did not first win its catalogue cell.

8 Artifacts and references

- Saved critique: `~/Downloads/Panoptic-carry.pdf / .html` (X article, @Panoptic_xyz, 2026-08-27, https://x.com/Panoptic_xyz/status/2093081191427805570).
- Panoptic research corpus (public): "Panoptic Solves LVR", "Reverse Gamma Scalping", "IV & Put Wing Premia", "Derivatives Solve the Uniswap Doom Loop" – <https://panoptic.xyz/research/>.
- LVR: Milionis, Moallemi, Roughgarden, Zhang, "Automated Market Making and Loss-Versus-Rebalancing", arXiv:2208.06046 (the $\sigma^2/8$ result).
- am-AMM: Adams et al., "am-AMM: An Auction-Managed Automated Market Maker", arXiv:2403.03367.
- Uniswap v4: core repo + hooks/dynamic-fee documentation.
- Hyperliquid: HIP-3 (builder-deployed perp markets) documentation.
- Lineage: Artemis Capital (C. Cole), "The Allegory of the Hawk and the Serpent" – the carry/convexity yin-yang framing Panoptic transplants.

- Parent documents: `REBALANCING-EQ.org` (the catalogue and its waves), `BASKET-REDESIGN.md` / `BASKET-REBALANCE.md` (custody + director), `alberta-buck-operations.org` (the ops desk and the profitability-vs-stabilization law), `alberta_buck/sim/EQUILIBRIUM.md` (the campaign and the controller).